Token-Share-Locked English–Tagalog Mixtures after a Fresh Tagalog Parent:

A Preregistered Trade-Off Study with nanochat

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21 August 2026

Abstract

After a newly trained Tagalog nanochat parent passes a filed eligibility gate, does a prospectively frozen mixture whose source-content token share is locked at $q_{\mathrm{TL}} = 0.50$ produce lower Tagalog held-out bits-per-byte (BPB) than pure English continuation and lower English held-out BPB than extra Tagalog continuation? This question was preregistered as AsPredicted #307591 after P3 unblinding and before any P4 tokenizer copy, parent, child, validation BPB, or test BPB existed. P4 is a post-P3, prospectively preregistered exposure-matched mixture trade-off study. It does not amend AsPredicted #306780, #306935, or #307342, and C3 is not P3 B3. A fresh Tagalog parent was trained from WikiText-TL-39 train documents on NVIDIA CUDA (A40) with the carry-forward P3 tokenizer, sequence length $T = 2048$, and Karpathy's nanochat trainer pinned at commit `92d63d4`. After P0-T PASS, the d20 checkpoint was frozen as immutable C0. Matched d20 continuations compared C1 extra Tagalog, C2 pure English, and C3 the frozen token-share mix for $D_{\mathrm{phase2}} =$ 19,267,584 tokens ($N = 294$). Co-primary estimands are $R_{\mathrm{TL}} = \mathrm{TL}(C3) - \mathrm{TL}(C2) \leq -0.01$ and $A_{\mathrm{EN}} = \mathrm{EN}(C3) - \mathrm{EN}(C1) \leq -0.01$. The Gate X release reports $R_{\mathrm{TL}} = -1.316637$ (observed) and $A_{\mathrm{EN}} = -1.375277$ (observed). One authorized C3-only secondary test event yielded English 1.513698 and Tagalog 1.202140 BPB; C1 and C2 were not tested. One seed; no confidence interval or population claim. The contribution is a locked token-share trade-off measurement in this apparatus, not a confirmation of P3, not "P3 B3 fixed," and not a general bilingual optimum.

Keywords: Tagalog retention; English acquisition; token share; continual pretraining; WikiText-TL-39; WikiText-103; bits-per-byte; nanochat; preregistration; post-P3.

arXiv categories: cs.CL, cs.LG.

Registration and deposits (P4): AsPredicted #307591 (https://aspredicted.org/if84km.pdf); ResearchBox #8869 (https://researchbox.org/8869; peer-review passcode `ABMSAN`; FOR PEER REVIEW, not Make Public); AsCollected #2471 (`NANOCHAT-FILIPINO-P4`); study record https://github.com/pageman/nanochat-filipino/tree/main/docs/p4; Hub https://huggingface.co/pageman/nanochat-filipino-p4-token-share-mix. Run ID: `p4-20260821T060032Z-92d63d4`.

1 Introduction

Pajo [1] measured equal-exposure Tagalog BPB across nanochat depths on WikiText-TL-39 (AsPredicted #306780). Pajo [2] asked the forward continual-pretraining question from an

English parent (AsPredicted #306935). Pajo [3] asked the reverse from a fresh Tagalog parent, with a 50/50-*document* B3 arm (AsPredicted #307342). P3 Gate X (20 August 2026) left a document/byte-share ambiguity: document-balanced B3 is not token-share balanced under a shared tokenizer.

P4 was designed after those released P3 findings were known. It is therefore a prospectively preregistered **post-P3** exposure-matched mixture trade-off study [4]. It is not an amendment, correction, rescue, replication, or reinterpretation of P3. It is not "P3 B3 fixed." C3 is a newly constructed token-share mixture; C3 is not P3 B3.

The confirmatory question, locked before any P4 confirmatory BPB existed, is:

> After a newly trained P4 Tagalog parent passes P0-T, freeze d20 as C0 and continue it for one shared phase-two token budget on C1 extra Tagalog, C2 pure English, and C3 a mixture whose P4-tokenizer-encoded source-content token share is locked at $q_{\mathrm{TL}} = 0.50$. Do $R_{\mathrm{TL}} = \mathrm{TL}(C3) - \mathrm{TL}(C2) \leq -\delta$ and $A_{\mathrm{EN}} = \mathrm{EN}(C3) - \mathrm{EN}(C1) \leq -\delta$ both hold at $\delta = 0.01$?

This paper contributes: (i) a locked token-share mixture constructed after tokenizer freeze and before any P4 BPB; (ii) a fresh Tagalog parent with P0-T PASS; (iii) six child validation cells plus descriptive C0 English, sealed before one C3-only secondary test touch; (iv) a Gate X release in which both co-primary criteria were observed in this one-seed apparatus.

2 Related work

Cruz and Cheng constructed WikiText-TL-39 [5]. Sequential interference is classical [11, 12]. LLM continual-pretraining work often continues an English or multilingual base [9, 10, 8]. P4 stays inside WikiText-family text, the nanochat depth dial, and BPB rather than CORE or chat [6]. Equal token budgets follow the family so stream identity is not confounded with compute [13, 14, 15]. WikiText established clean Wikipedia LM evaluation in English [7]. BPB remains comparable across vocabularies [16, 17]. Official P4 BPB is mean token NLL divided by $\ln 2$ times mean UTF-8 bytes per token, full split, $T = 2048$, BOS-best-fit. In-loop trainer BPB is not confirmatory. Preregistration reduces undisclosed flexibility [18, 19].

3 Methods

3.1 Registration and authority

AsPredicted #307591 governs [4].

- PDF: https://aspredicted.org/if84km.pdf
- PDF SHA-256:
  `463b29fcff8d7c8099790325fa19d6bcf9ee29f64424c373a380566a6fe9011c`
- Protocol SHA-256 at filing:
  `22c28f2bc632f132d9c95bbbcc9d1facbddf0b6b821445487e451c472ea58d4b`

ResearchBox #8869 is the deposit (https://researchbox.org/8869; peer-review passcode `ABMSAN`; the box remains FOR PEER REVIEW and is not made public by this paper). AsCollected #2471 (`NANOCHAT-FILIPINO-P4`) records public-data provenance. Authority: filed PDF

$\gg$ protocol $\gg$ `LOCK.json` $\gg$ dated deviations $\gg$ this manuscript. The study does not amend #306780, #306935, or #307342. C3 is not P3 B3.

3.2 Sources, tokenizer, and parent

Tagalog train/val/test documents are the frozen WikiText-TL-39 P1.1-eligible splits. English train/val/test documents are the frozen WikiText-103 P2-eligible splits. The tokenizer is the carry-forward P3 pair: `tokenizer.pkl` `04436b854e0841025a3dd2b46baaeeea07a7ccc252e9f99a19171306f00bc5a8` and `token_bytes.pt` `a5dbc1c88f6292696108263072d77115718cc2d8357f7ad4859adfa517cc2132`. P1.1/P2/P3 weights were not loaded as parents. The parent was trained from random initialization on Tagalog train only (Gates I then Q). Confirmatory GPU gates used NVIDIA CUDA A40.

3.3 P0-T and C0 freeze

P0-T compared full-split Tagalog validation BPB of d8 and d20 TL0 against an untrained initialization and a byte-unigram floor at $\delta_{\mathrm{P0T}} = 0.01$. Official P0-T status is CUDA-only. After PASS, d20 was copied to an immutable C0 directory with `additional_train_tokens`=0. Smoke (d4) and d8 checkpoints are not C0.

3.4 Children and C3 construction

C1, C2, and C3 each continued C0 for $N = 294$ steps, $B = 65{,}536$, $D_{\mathrm{phase2}} = 19{,}267{,}584$ tokens, seed 42, serial R→S→T. C1 used Tagalog train only. C2 used English train only. C3 was packed after tokenizer freeze (operational F then E) to exact source-content token quotas 9,633,792 Tagalog and 9,633,792 English ($q_{\mathrm{TL}} = 0.50$). Byte share is descriptive, not the exposure clock. Mix-manifest SHA-256: `f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107`.

3.5 Lockbox, analysis, and tests

Validation BPB was lockboxed until Gate X. Gate U sealed six child cells plus descriptive C0 English with test-access count 0. Policy A authorized one C3-only test event after U (Gate V). C1 and C2 were not tested. Contrasts use the sealed validation cells; tests are secondary. Reporting precision is six decimal places. Equality at $-\delta$ counts. No composite score, confidence interval, or p-value.

4 Results

P0-T status is **PASS**. Depth-8 Tagalog val BPB was 1.179921 versus untrained 3.289109 and byte-unigram 4.453225. Depth-20 Tagalog val BPB was 1.172168 versus untrained 3.289106 and the same unigram floor. C0 SHA-256 matches the Gate I d20 checkpoint `34e069646be4158979809c023691188439047d6cbee08a141db432c78bcf02e2`.

Table 1 reports the six child validation cells and descriptive C0 English. Co-primary contrasts from the Gate U seal are

$$R_{\mathrm{TL}} = \mathrm{TL}(C3) - \mathrm{TL}(C2) = -1.316637 \leq -0.01 \quad \text{(observed)}, \tag{1}$$
$$A_{\mathrm{EN}} = \mathrm{EN}(C3) - \mathrm{EN}(C1) = -1.375277 \leq -0.01 \quad \text{(observed)}. \tag{2}$$

C0 English (2.615645) is descriptive and excluded from the contrasts.

Table 1: Full-split validation BPB after terminal d20 checkpoints. Six-decimal reporting as filed. C0 English is descriptive.

Arm	Tagalog val BPB	English val BPB
C0 (frozen parent)	—	2.615645
C1 extra Tagalog	0.785486	2.878106
C2 pure English	2.517909	1.333106
C3 token-share mix	1.201273	1.502828

Under this preregistered token-share-locked mixture, P4 observed lower Tagalog BPB than pure English continuation and lower English BPB than extra Tagalog continuation in the stated apparatus. One seed; no CI/p-value/population claim; C3 is not P3 B3; P4 does not amend #307342.

A measured reduction in the P3-style relative Tagalog retention cost within this frozen P4 apparatus, not a general mitigation of catastrophic forgetting.

One authorized C3-only secondary test event reported English test BPB 1.513698 and Tagalog test BPB 1.202140. Tests do not alter sealed contrasts and are not co-primary.

5 Discussion

C3 sits between C1 and C2 on both languages, as a token-share trade-off rather than a claim that $q_{\mathrm{TL}} = 0.50$ is optimal. Byte share of the packed mix is descriptive; 50/50 source-content tokens is not 50/50 documents and is not 50/50 UTF-8 bytes. The result is a pattern observed in one seed under a frozen P4 apparatus. It does not independently confirm P3. It does not show that P3 B3 was "fixed." It does not establish a universal mixture law.

6 Limitations

One seed; one tokenizer; WikiText-family text only; no chat, SFT, CORE, or multi-seed population inference. Confirmatory GPU work used a rented NVIDIA A40. Raw test JSONL remains restricted. Hub weight release, if any, must ship C0+C1+C2+C3 together with tokenizer and meta; C3 alone is prohibited. Byte-balanced mix is a separate filing (P6-B). Multi-seed is a separate filing (P5).

A technical incident at Gate V (missing English test JSONL on the GPU host) failed before any test score. Files were restored; V was rerun once from the U seal with test-access still 0 at U. Frozen English train JSONL contains intra-split duplicate rows (28,472 rows / 28,232 unique); they were not dropped. Cross-split overlap was 0.

7 Deviations

No protocol stop. No break-glass. No additional confirmatory validation or test eval after Gate X. The V JSONL miss is a technical incident, not a science change.

Availability

- AsPredicted #307591: https://aspredicted.org/if84km.pdf
- ResearchBox #8869: https://researchbox.org/8869
  (peer-review passcode `ABMSAN`; FOR PEER REVIEW; not Make Public)
- AsCollected #2471 (`NANOCHAT-FILIPINO-P4`)
- Study record (GitHub):
  https://github.com/pageman/nanochat-filipino/tree/main/docs/p4
- Code and audit trees (GitHub):
  https://github.com/pageman/nanochat-filipino
  P4-only: `scripts/p4/`, `docs/papers/p4-token-share-mix/`, `docs/run-cards/p4/`, `results/p4/`, `docs/hub/p4-token-share-mix/`
- Weights (Hugging Face Hub):
  https://huggingface.co/pageman/nanochat-filipino-p4-token-share-mix
  (C0+C1+C2+C3 together; never C3 alone; not on P1.1/P2/P3 Hub IDs).

Held-out test text is not redistributed. Host credentials are not published.

Ethics

Public Wikipedia-derived corpora. No human-subjects experiment. The ResearchBox remains passcode-protected for peer review and is not made public by this paper.

Acknowledgements

Thanks to Manus 1.6 and Cursor.com(Auto) for the drafting, formatting, and solutioning of the paper. WikiText-TL-39 is due to Cruz and Cheng [5]. WikiText-103 is due to Merity et al. [7]. nanochat is due to Karpathy [6]. Errors remain the author's.

Funding

No dedicated grant. GPU time (Runpod A40) was paid by the author.

A Selected hashes

- Pin / nanochat: `92d63d4e8bb4df75c3b71618f31ddde2378b2bcd`

- C0 / I d20: `34e069646be4158979809c023691188439047d6cbee08a141db432c78bcf02e2`
- C1: `87b9f55146de72dd6ae53598b9aea8d99079ff0f9492b7f9ea4fdce550664c55`
- C2: `0787aed0f13a0ab3ec144baf6802b144a18412780a2d00a64ca7adcb67a4a375`
- C3: `eef9a4e11c4840ac036d42c3bf4d87a2139ea1fa5809e1c756df2770fe0609f3`
- U seal: `5c7287752ba1abb39245acab43b9917ea9e089c0309959ec24990015e1ad580f`
- Evaluator: `9afebdb405aaac0bb4287051d9b6f5d16f56d6dd9269a1e6c2c5df29becbced1`
- Protocol: `22c28f2bc632f132d9c95bbbcc9d1facbddf0b6b821445487e451c472ea58d4b`
- Gate bible: `b389b70e0b8e3af869e8dea314b1c7c6b91df313e49d1bf11d9d07961b4a5a42`
- Addendum: `f056a6f75c73a4d8dc3401ba8d7219d406aa7e498e5b0799d3d0373f9f74c216`
- AsPredicted PDF: `463b29fcff8d7c8099790325fa19d6bcf9ee29f64424c373a380566a6fe9011c`

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